Delays and Filters

Intro to Audio Tech 1
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Lesson Outline

- Delay Review
- Filters
- Up Next…

Filters

Simplest Possible Filter

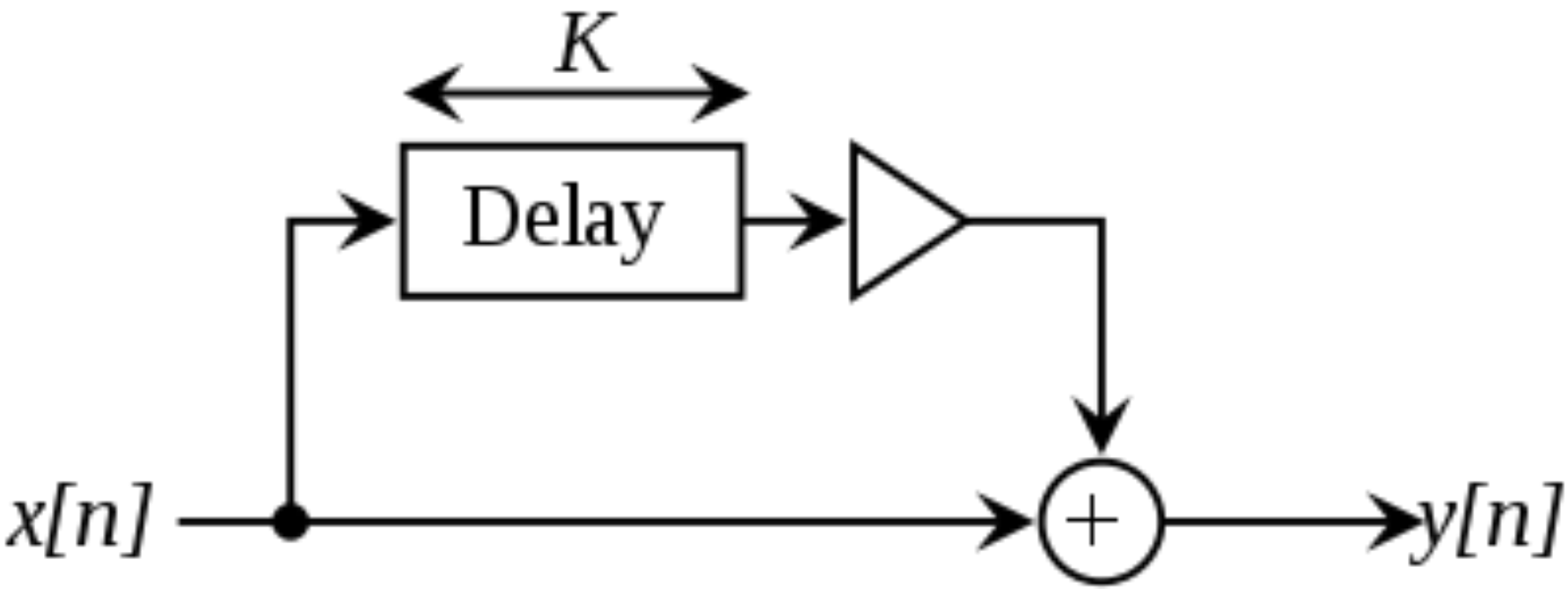
- Signal x added to some scaled delayed version of itself

$$y_t = x_t + a_1 x_{t-\tau}$$

$$y[n] = \mathrm{x[n]} + a_1 x[n-1]$$

- We can ”plug-in” numbers sample by sample

Simplest Possible Filter – Block Diagram



Impulse Response

- We can send an impulse through our filter to determine how it reacts in the **time domain**
- A perfect impulse is a short burst signal containing all frequencies
  - This looks like a signal of [1, 0, 0, 0, 0, 0…]
- For LTIs, the impulse response fully characterizes the filter
- For the "Simplest Possible Filter":

$$y[n] = \mathrm{x[n]} + a_1 x[n-1]$$

$$\mathrm{h}[n] = \{1, 1, 0, 0, 0, 0..\}$$

Filters in the Frequency Domain

Frequency Response

- We can send the same unit impulse through a filter to determine how it acts in the **frequency domain**
- However, this time instead of looking at how the input changes, we are looking at how the spectrum changes
- The unit impulse contains a flat spectrum of all frequencies, so we find the frequency response to determine how those frequencies are attenuated

Rewriting the Simplest Possible Filter

- We can rewrite the filter using complex phasors
- If we delay the phasor $e^{j\omega t}$ by τ…
- A delay of τ seconds multiplies the phasor by the complex factor $e^{-j\omega\tau}$

$$e^{j\omega(t-\tau)} = e^{-j\omega\tau}\ e^{j\omega t}$$

- Remember this factor:
  - This does not depend on time
  - Only depends on the delay time and frequency

Simplest Possible Filter

Algebraic Representation: $y_t = x_t + a_1 x_{t-\tau}$

Phasor Representation: $y_t = e^{j\omega t} + a_1 e^{j\omega(t-\tau)}$

Phasor (simplified): $y_t = e^{j\omega t}(1 + a_1 e^{-j\omega\tau})$

- What does this tell us?
  - The output of the filter is the phasor at frequency ω
  - The effect of the filter is to multiply it by the complex function

Simplest Possible Filter

Frequency Response:

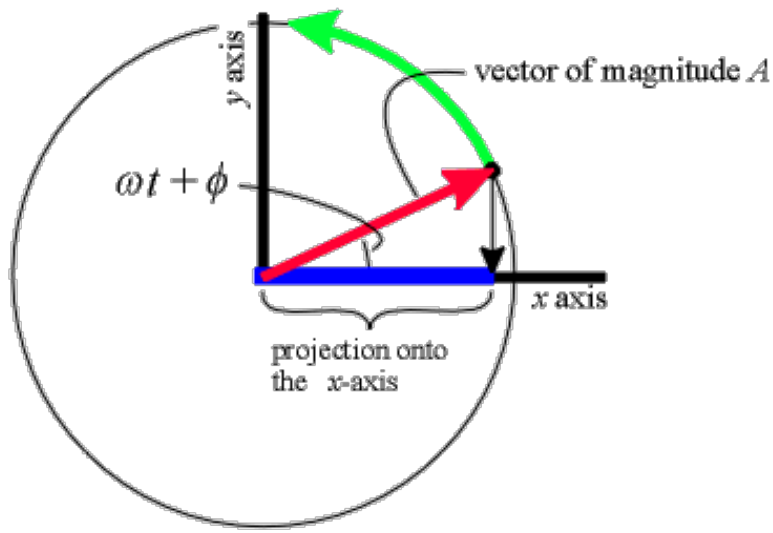
$$H(\omega) = \left(1 + a_1 e^{-j\omega\tau}\right)$$

- Depends on frequency and fixed delay
  - NOT on time
- In polar form:

$$H(\omega) = |H(\omega)| e^{j\theta(\omega)}$$

Frequency Response

- $H(\omega) = |H(\omega)|e^{j\theta(\omega)}$
- This tells us both
  - The frequencies outputted amplitude/power because of the magnitude response
  - The frequencies shifted phase because of the phase response



Impulse vs Frequency Response

- Impulse Response
  - The filter's response to a unit impulse – how it characterizes that impulse
  - "How are the outputted samples affected"
  - Time domain
- Frequency Response
  - The filter's response to a unit impulse
  - How does the system respond to specific frequencies
  - Frequency domain